Final Exam

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Problem 1

Cholesterol is a type of lipid, a fat-like substance that's essential for many bodily functions. The Blood fat measure can serve as a good estimate for the cholesterol. The "BF" dataset provides the observations of 25 adults. With this dataset, a predictive model can be performed to estimate the Blood Fat Y

$$\hat{Y} = \beta_0 + \beta_1 X_i 1 + \beta_2 X_i 2 + \epsilon$$

```r
Weight <- read.csv("~/R for Statistics and Data Science (STAT-5521G)/Weight.txt", sep="")
```

Construct the linear regression model and report the R output of the summary.

```r
muti_lr <- lm(Blood_Fat ~ ., data = Weight)

summary(muti_lr)
```

```
##
## Call:
## lm(formula = Blood_Fat ~ ., data = Weight)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -69.570 -30.374  -5.449  28.626  89.170
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  77.9825    52.4296   1.487    0.151
## Weight        0.4174     0.7288   0.573    0.573
## Age           5.2166     0.7572   6.889 6.43e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 44.11 on 22 degrees of freedom
## Multiple R-squared:  0.7056, Adjusted R-squared:  0.6788
## F-statistic: 26.36 on 2 and 22 DF,  p-value: 1.443e-06
```

We fitted a linear regression model using ordinary least squares (OLS) to predict **Blood Fat** levels from **Weight** and **Age** (`Blood_Fat ~ Weight + Age`). The model explains a statistically significant and substantial proportion of the variance in blood fat levels:

- $R^2 = 0.71$, adjusted $R^2 = 0.68$
- $F(2, 22) = 26.36,\ p < .001$

Intercept

The model's intercept, corresponding to Weight = 0 and Age = 0, was estimated at **77.98**, with a 95% confidence interval of $[-30.75, 186.71]$. This was not statistically significant: - $t(22) = 1.49,\ p = 0.151$

Weight

The effect of Weight on Blood Fat was positive but not statistically significant: - $\beta = 0.42$, 95% CI $[-1.09, 1.93]$ - $t(22) = 0.57,\ p = 0.573$
- Standardized beta = **0.07**, 95% CI $[-0.18, 0.32]$

Age

The effect of Age on Blood Fat was statistically significant and positive: - $\beta = 5.22$, 95% CI $[3.65, 6.79]$ - $t(22) = 6.89,\ p < .001$
- Standardized beta = **0.82**, 95% CI $[0.57, 1.07]$

1.2. Report on the fitted linear regression model.

$$\hat{BloodFat} = \beta_0 + \beta_1 Weight + \beta_2 Age$$
$$\hat{BloodFat} = 77.9825 + 0.4174(Weight) + 5.2166(Age)$$

1.3. Report the R ANOVA table of the model (the output of the ANOVA in R).

```r
anova(muti_lr)
```

```
## Analysis of Variance Table
##
## Response: Blood_Fat
##           Df Sum Sq Mean Sq F value   Pr(>F)
## Weight     1  10232   10232  5.2585  0.03177 *
## Age        1  92339   92339 47.4571 6.43e-07 ***
## Residuals 22  42806    1946
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

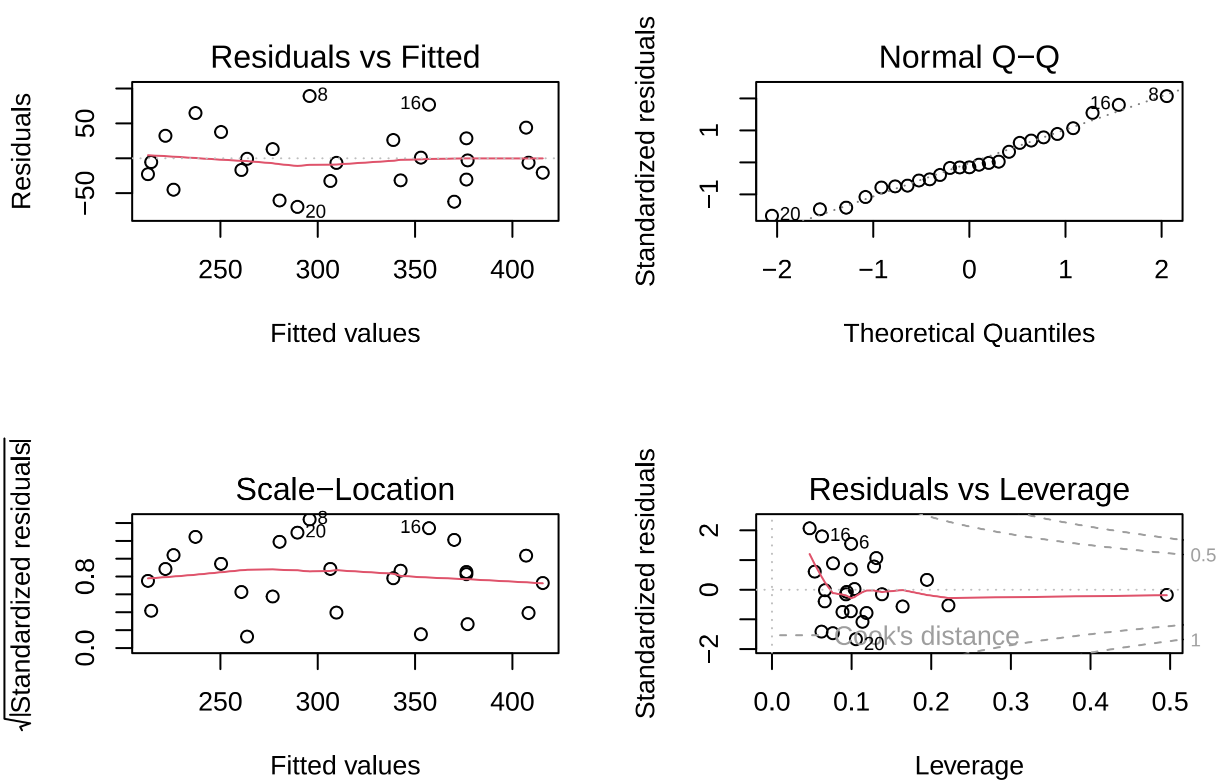
- **Main Effect of Weight**:
  There is a **statistically significant** and **large** effect of **Weight** on blood fat levels, $F(1, 22) = 5.26$, $p = 0.032$.
  The **partial Eta-squared** (η_p^2) is **0.19** (95% CI: [0.01, 1.00]), indicating that approximately 19% of the variance in blood fat levels is explained uniquely by Weight.

- **Main Effect of Age**:
  There is a **highly significant** and **large** effect of **Age**, $F(1, 22) = 47.46$, $p < 0.001$. The **partial Eta-squared** (η_p^2) is **0.68** (95% CI: [0.48, 1.00]), suggesting that Age uniquely explains about 68% of the variance in blood fat levels.

Overall, Both **Weight** and **Age** have statistically significant and practically meaningful effects on blood fat levels, with Age contributing a substantially larger portion of the explained variance.

1.4. Check whether the assumptions for the linear regression model are satisfied.

```r
# Change the panel layout to 2 x 2 (to look at all 4 plots at once)
par(mfrow = c(2, 2))
plot(muti_lr)
```




```r
shapiro.test(residuals(muti_lr))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(muti_lr)
## W = 0.97054, p-value = 0.6588
```

1.5 Interpret what the value of β_1 means

In this model, β_1 represents the estimated effect of **Weight** on **Blood Fat**, holding **Age** constant. The estimated value of $\beta_1 = 0.42$ means that, on average, for each additional unit of weight (e.g., 1 kg), blood fat is expected to increase by **0.42 units**, assuming age remains constant. However, this effect is **not**

statistically significant ($p = 0.573$), and the 95% confidence interval $[-1.09,\ 1.93]$ includes zero. This suggests that the observed relationship between weight and blood fat could be due to random chance, and there is **no strong evidence** that weight is a meaningful predictor of blood fat in this model.

1.6. According to the output, is there a significant linear relationship between the blood fat and the two independent variables? Provide a summary of the hypothesis test that was performed using the F-test statistics, clearly state the Hypothesis, the test statistics, the p-value, the decision rule and the conclusion.

1.6 Test for Overall Significance using the F-Test

We conducted an F-test to determine whether there is a statistically significant linear relationship between **Blood Fat** (dependent variable) and the two independent variables (**Weight** and **Age**).

Hypotheses

- **Null Hypothesis** (H_0): $\beta_1 = \beta_2 = 0$
  (i.e., neither Weight nor Age has a linear effect on Blood Fat)
- **Alternative Hypothesis** (H_1): At least one $\beta \neq 0$
  (i.e., at least one predictor has a significant linear relationship with Blood Fat)

Test Statistic

- **F(2, 22) = 26.36**

P-value

- **p < .001**

Decision Rule

- If *p-value* < 0.05, reject the null hypothesis H_0

Conclusion

Since the p-value is less than 0.001, which is well below the 0.05 threshold, we **reject the null hypothesis**. This indicates that there is a **statistically significant linear relationship** between Blood Fat and the set of predictors (Weight and Age) as a whole.

However, while the overall model is significant, the **individual effect of Weight was not statistically significant**, whereas **Age was a strong and significant predictor**.

1.7 According to the output perform the individual t-tests for significance of each independent variable.

1. Weight

- **Null Hypothesis** $H_0 : \beta_1 = 0$

- **Alternative Hypothesis** $H_1 : \beta_1 \neq 0$
- Estimated Coefficient: **0.42**
- 95% CI: [**-1.09, 1.93**]
- **t(22) = 0.57**, **p = 0.573**

Since the p-value (0.573) is **greater than 0.05**, we **fail to reject the null hypothesis**. There is **no statistically significant evidence** that **Weight** is associated with Blood Fat in this model.

2. Age

- **Null Hypothesis** $H_0 : \beta_2 = 0$
- **Alternative Hypothesis** $H_1 : \beta_2 \neq 0$
- Estimated Coefficient: **5.22**
- 95% CI: [**3.65, 6.79**]
- **t(22) = 6.89**, **p < 0.001**

Since the p-value is **less than 0.001**, we **reject the null hypothesis**. There is **strong statistically significant evidence** that **Age** is positively associated with Blood Fat.

1.8. Find an estimate for the expected blood fat level if the weight of the person is 88 kg and the age of the same person is is 58 years.

We use the fitted regression model to estimate the expected blood fat level:

$$\hat{Y} = 77.98 + 0.42 \times 88 + 5.22 \times 58$$

$$\hat{Y} = 77.98 + 36.96 + 302.76 = 417.70$$

Estimated Blood Fat = 417.70 units

1.9. The following model was fit:

$$\hat{Y} = \beta_0 + \beta_1 X_i 1 + \beta_2 X_i 2 + \beta_3 X_i 1 X_i 2 + \epsilon$$

Provide the R summary of the full regression model

```
full_muti_lr <- lm(Blood_Fat ~ Weight*Age, data = Weight)

summary(full_muti_lr)
```

```
##
## Call:
## lm(formula = Blood_Fat ~ Weight * Age, data = Weight)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -68.910 -26.925  -7.654  30.275  89.961
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 141.67498  161.54291   0.877    0.390
## Weight       -0.53587    2.39943  -0.223    0.825
## Age           3.23690    4.80082   0.674    0.508
## Weight:Age    0.02910    0.06964   0.418    0.680
##
## Residual standard error: 44.96 on 21 degrees of freedom
## Multiple R-squared:  0.708,  Adjusted R-squared:  0.6663
## F-statistic: 16.97 on 3 and 21 DF,  p-value: 7.899e-06
```

1.10. Provide the output of the ANOVA table in R for the full model

```r
anova(full_muti_lr)
```

```
## Analysis of Variance Table
##
## Response: Blood_Fat
##            Df Sum Sq Mean Sq F value    Pr(>F)
## Weight      1  10232   10232  5.0612   0.03532 *
## Age         1  92339   92339 45.6765 1.099e-06 ***
## Weight:Age  1    353     353  0.1746   0.68033
## Residuals  21  42453    2022
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

1.11. Report the ANOVA output in R for the full vs. reduced model.

```r
anova(muti_lr,full_muti_lr)
```

```
## Analysis of Variance Table
##
## Model 1: Blood_Fat ~ Weight + Age
## Model 2: Blood_Fat ~ Weight * Age
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     22 42806
## 2     21 42453  1    352.88 0.1746 0.6803
```

1.12 In testing whether β_3 can be dropped from the model, state the null and alternative hypotheses.

Null Hypothesis (H_0): $\beta_3 = 0$

Alternative Hypothesis (H_a): $\beta_3 \neq 0$

1.13. Calculate the test statistic to be used to test the hypotheses in 1.11 What degrees of freedom are used for this test?

$$F = \frac{SS_{ER} - SS_F}{df_R - df_F} \bigg/ \frac{SS_F}{df_F}$$

$$F = \frac{42806 - 42453}{22 - 21} \bigg/ \frac{42453}{21}$$

$$F = \frac{352.88}{1} \bigg/ 2021.57 \approx 0.1746$$

- Numerator degrees of freedom: **q = 1**
- Denominator degrees of freedom: **dfres = 21**

1.14. How will you interpret the output in 1.10? State your conclusion

Since the **P-value (0.6803)** > 0.05 level of significance, we fail to reject the **Null hypothesis** and conclude that $\beta_3 = 0$ and it needed to be dropped in the model.

1.15. Fit a first-order regression model to the data for the two predictor variables. Obtain the externally studentized residuals and identify any outlying cases. State your conclusion

```r
muti_lr <- lm(Blood_Fat ~ ., data = Weight)

# Obtain externally studentized residuals
ext_resid <- rstudent(muti_lr)
# View residuals with original data
Weight$Externally_Studentized_Residuals <- ext_resid
# Identify outliers (typically abs(residual) > 2)
Weight$Outlier <- abs(ext_resid) > 2
# Print results
print(Weight)
```

```
##    Weight Age Blood_Fat Externally_Studentized_Residuals Outlier
## 1      84  46       354                       0.02329851   FALSE
## 2      73  20       190                      -0.55594699   FALSE
## 3      65  52       405                       0.67519737   FALSE
## 4      70  30       263                      -0.01594765   FALSE
## 5      76  57       451                       1.07254059   FALSE
## 6      69  25       302                       1.60229482   FALSE
## 7      63  28       288                       0.88405122   FALSE
## 8      72  36       385                       2.25521995    TRUE
## 9      79  57       402                      -0.15038549   FALSE
## 10     75  44       365                       0.60132838   FALSE
```

```
## 11      27  24       209                          -0.17009729   FALSE
## 12      89  31       290                           0.32556034   FALSE
## 13      65  52       346                          -0.71739780   FALSE
## 14      57  23       254                           0.77581674   FALSE
## 15      59  60       395                          -0.52053463   FALSE
## 16      69  48       434                           1.90330054   FALSE
## 17      60  34       220                          -1.44858847   FALSE
## 18      79  51       374                          -0.06982276   FALSE
## 19      75  50       308                          -1.50698734   FALSE
## 20      82  34       220                          -1.74308314   FALSE
## 21      59  46       311                          -0.74198552   FALSE
## 22      67  23       181                          -1.08627335   FALSE
## 23      85  37       274                          -0.77708915   FALSE
## 24      55  40       303                          -0.15360306   FALSE
## 25      63  30       244                          -0.38584879   FALSE
```

An observation is typically considered an outlier if its studentized residual exceeds ± 2. Only one observation was identified as an outlier. This observation may disproportionately affect model estimates.

Obtain the diagonal element of the hat matrix (the leverage). Identify any outlying X observations using the cutoff value at 2p/n

```r
# Compute leverage values (diagonal of the hat matrix)
leverage <- hatvalues(muti_lr)
# Determine p (number of parameters including intercept) and n (sample size)
p <- length(coef(muti_lr)) # includes intercept
n <- nrow(Weight)
# Calculate cutoff value for high leverage
cutoff <- 2 * p / n
# Identify high leverage observations
Weight$Leverage <- leverage
Weight$High_Leverage <- leverage > cutoff
# View results
print(paste("Leverage cutoff is:", round(cutoff, 4)))
```

```
## [1] "Leverage cutoff is: 0.24"
```

```r
subset(Weight, High_Leverage == TRUE)
```

```
##    Weight Age Blood_Fat Externally_Studentized_Residuals Outlier  Leverage
## 11     27  24       209                        -0.1700973   FALSE 0.4957708
##    High_Leverage
## 11          TRUE
```

Points with leverage > 2p/n are considered high leverage. Observations 11 was identified as high leverage points based on the hat matrix diagonals. This is the data point with unusually extreme predictor value compared to the rest of the dataset. High leverage points have the potential to strongly influence the regression model, especially the fitted values.

1.17. Obtain the DFFITS, DFBETAS, and Cook's Distance values for assessing influential observations. Discuss your findings.

```
# DFFITS
dffits_vals <- dffits(muti_lr)
# DFBETAS
dfbetas_vals <- dfbetas(muti_lr)
# Cook's Distance
cooks <- cooks.distance(muti_lr)
# Thresholds:
dffits_cutoff <- 2 * sqrt(p / n)
dfbetas_cutoff <- 2 / sqrt(n)
influential_points <- which(abs(dffits_vals) > dffits_cutoff |
apply(abs(dfbetas_vals) > dfbetas_cutoff, 1, any) |
cooks > 1)
influential_points
```

```
##  6 20
##  6 20
```

Observations 6 and 20 appear consistently across multiple influence diagnostics (DFFITS,DFBETAS, and Cook's Distance). This consistency suggests that these points may have a disproportionate impact on your regression model's estimates, including coefficients and predictions.

1.18. Obtain the analysis of variance table that decomposes the regression sum of squares into extra sums of squares associated with X_2 with given X_1

```
# Fit the reduced model with only X1 (Weight)
model_reduced <- lm(Blood_Fat ~ Weight, data = Weight)

# Fit the full model with both X1 and X2 (Weight + Age)
model_full <- lm(Blood_Fat ~ Weight + Age, data = Weight)

# Compare the two models using ANOVA
anova(model_reduced, model_full)
```

```
## Analysis of Variance Table
##
## Model 1: Blood_Fat ~ Weight
## Model 2: Blood_Fat ~ Weight + Age
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1     23 135145
## 2     22  42806  1     92339 47.457 6.43e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The extra sum of squares associated with Age given Weight is 92339. The test statistic is highly significant ($F = 47.457$, $p < 0.001$), indicating that Age provides a significant improvement to the model after accounting for Weight.

1.19. Provide R output for Multicollinearity test.

```
library(car)
vif(model_full)
```

```
##   Weight      Age
## 1.061128 1.061128
```

Both Weight and Age have VIF values close to 1, which indicates no multicollinearity.

1.20. Provide interpretation for the R-squared and and what does it represents?

The formula for the coefficient of determination R^2 is:

$$R^2 = 1 - \frac{\mathrm{SS}_{\mathrm{res}}}{\mathrm{SS}_{\mathrm{tot}}}$$

```
summary(model_full)$r.squared
```

```
## [1] 0.7055503
```

R-squared: 0.7056

This means that approximately 70.56% of the variation in Blood_Fat can be explained by the predictors Weight and Age in the model. The remaining 29.44% of the variation is due to other factors not included in the model or random variation.

1.21. What is you conclusion about the current model

The current multiple linear regression model, which includes Weight and Age as predictors of Blood Fat, explains approximately **70.56%** of the variability in the response variable. The model shows a statistically significant improvement when Age is added **($\mathbf{F = 47.457, p < 0.001}$)**, indicating that both predictors contribute meaningfully to the explanation of Blood_Fat levels. There is no strong evidence of multicollinearity if VIF values remain below critical thresholds. Therefore, the current model is appropriate and provides a good fit to the data.

Problem 4

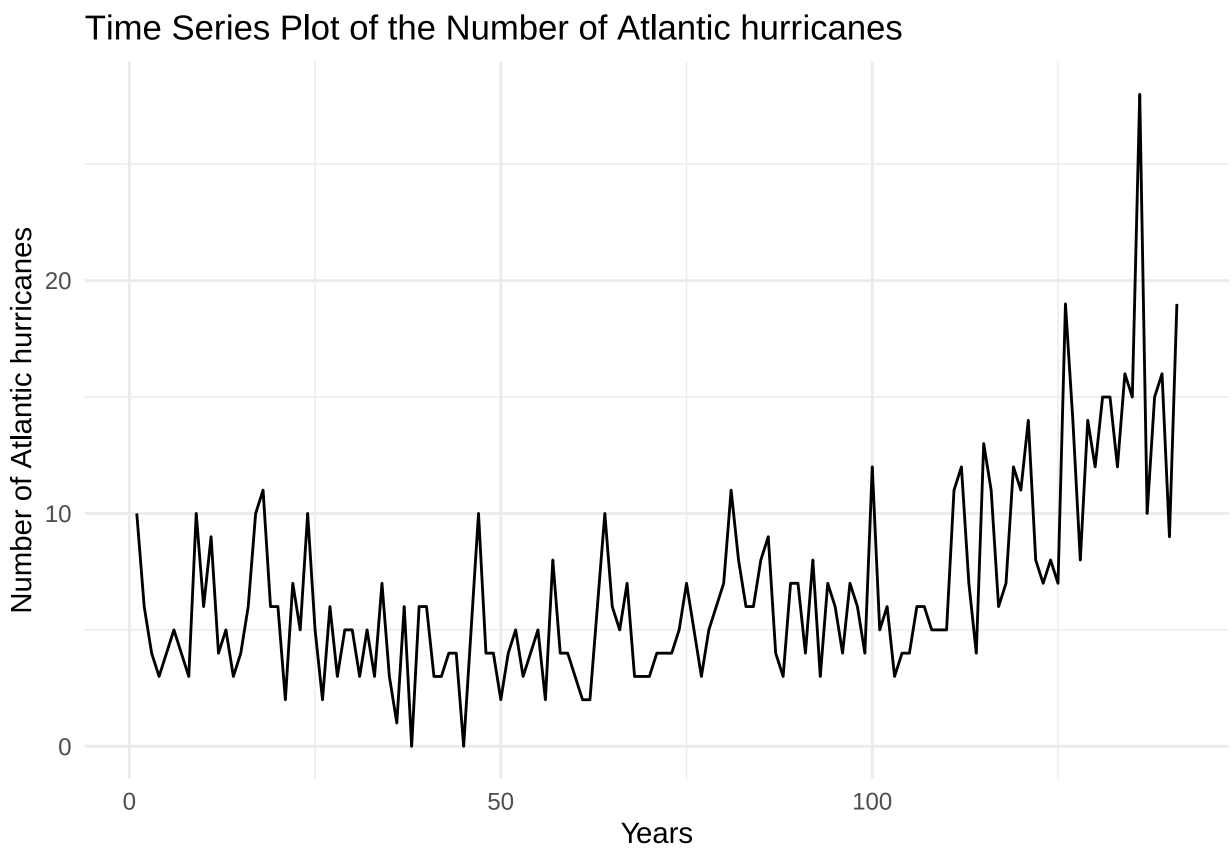
HURRICANES is a data set of two columns. Column 1 is the year, between 1870 and 2010, and column 2 is the number of Atlantic hurricanes. Using the best possible ARIMA model, according to your analysis, please provide the R output for the model and answer the following questions:

```
#Load Packages
library(lubridate)
library(forecast)
library(tseries)
library(ggplot2)


# Loading Hurrucanes data set
timedata <- read.csv("~/R for Statistics and Data Science (STAT-5521G)/timedata.txt", sep="")
```

```
timedata$Number_of_Hur <- ts(timedata$Number_of_Hur)

autoplot(timedata$Number_of_Hur) +
  ggtitle("Time Series Plot of the Number of Atlantic hurricanes") +
  labs(x = "Years", y = "Number of Atlantic hurricanes")+
  theme_minimal()
```

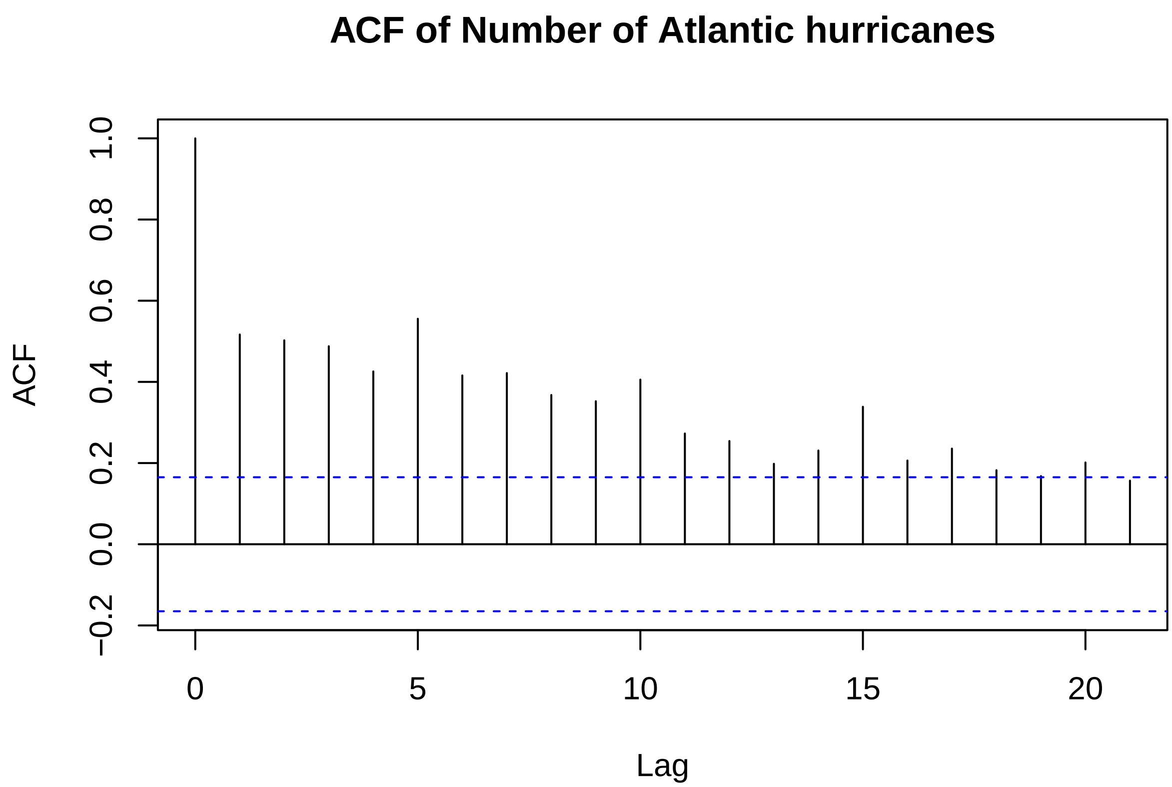


4.1. Does the data look stationary? Provide the R output. Do you need any transformation if you do provide an R output for it?

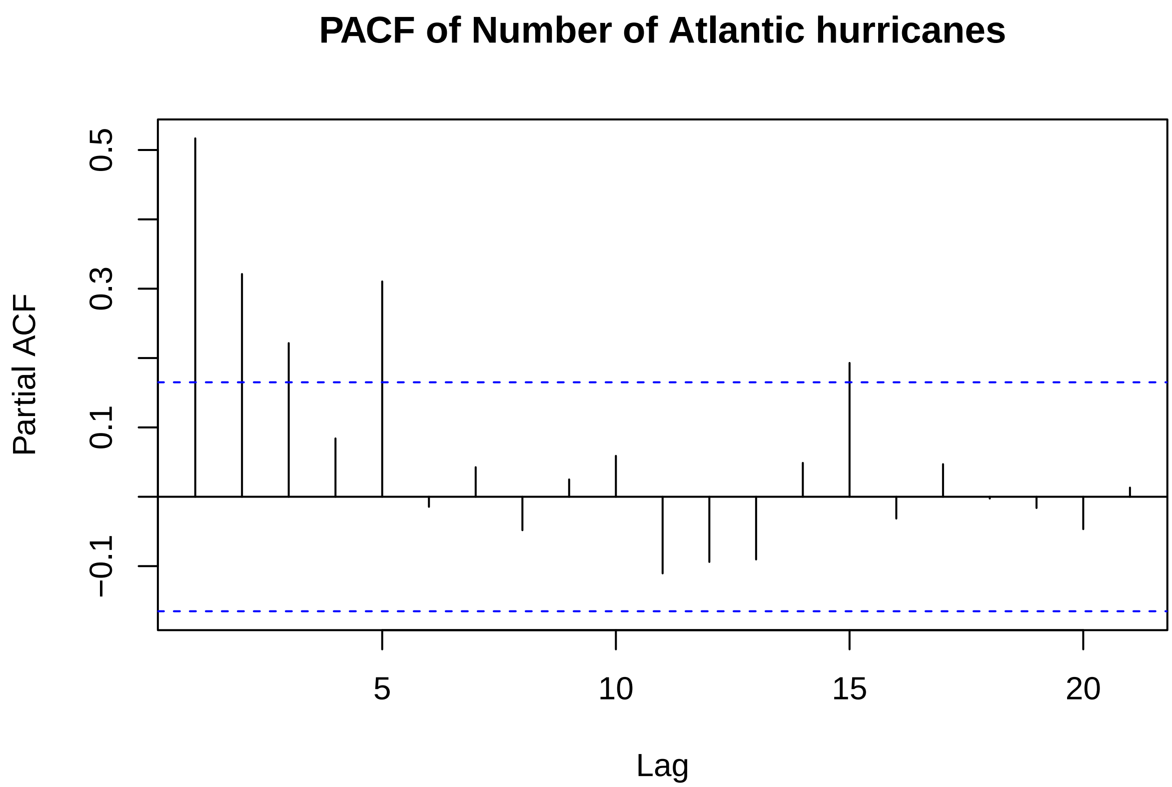
Null Hypothesis: The time series is non-stationary.

Alternative Hypothesis: The time series is stationary.

```
# Plot ACF and PACF
acf(timedata$Number_of_Hur, main = "ACF of Number of Atlantic hurricanes")
```




```r
pacf(timedata$Number_of_Hur, main = "PACF of Number of Atlantic hurricanes")
```




```r
# Perform Augmented Dickey-Fuller Test
adf.test(timedata$Number_of_Hur)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  timedata$Number_of_Hur
## Dickey-Fuller = -1.727, Lag order = 5, p-value = 0.6901
## alternative hypothesis: stationary
```

Decision and Conclusion: Since the p-value associated with the Dickey-Fuller test is greater than 0.05, we have all evidence to conclude that the data is non-stationary. Therefore, transformation and differencing of the data is inevitable.

4.2. Provide an R output for test for stationarity.

```r
#Transformation and Differencing
diff_timedata <- diff(timedata$Number_of_Hur)

# Test of Stationary
adf.test(diff_timedata)
```

```
##
```

```
##  Augmented Dickey-Fuller Test
##
## data:  diff_timedata
## Dickey-Fuller = -8.5417, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

Decision and Conclusion: Since the p-value associated with the Dickey-Fuller test is lesser than 0.05 level of significance, we have all evidence to conclude that the data is stationary. Therefore, we can go ahead to modeling our data.

4.3. What is the order of the ARIMA model? Provide the R output of how did you find it?

```r
model <- auto.arima(diff_timedata)

model
```

```
## Series: diff_timedata
## ARIMA(0,0,1) with non-zero mean
##
## Coefficients:
##           ma1    mean
##       -0.8341  0.0694
## s.e.   0.0395  0.0443
##
## sigma^2 = 9.417:  log likelihood = -355.22
## AIC=716.43   AICc=716.61   BIC=725.26
```

4.4. Report the theoretical model with the corresponding parameters.

$$\nabla Y_t = Y_t - Y_{t-1} = \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

Where: - Y_t is the original time series at time t - ∇Y_t represents the first difference - ε_t is the white noise (error term) - θ_1 is the MA(1) coefficient

$\theta_1 = -0.8341$

$c = 0.0694$

Final Model is: $Y_t = Y_{t-1} + 0.0694 + \epsilon_t - 0.8341\epsilon_{t-1}$

4.5. Provide an estimated value for the parameters, using R and the fitted model output.

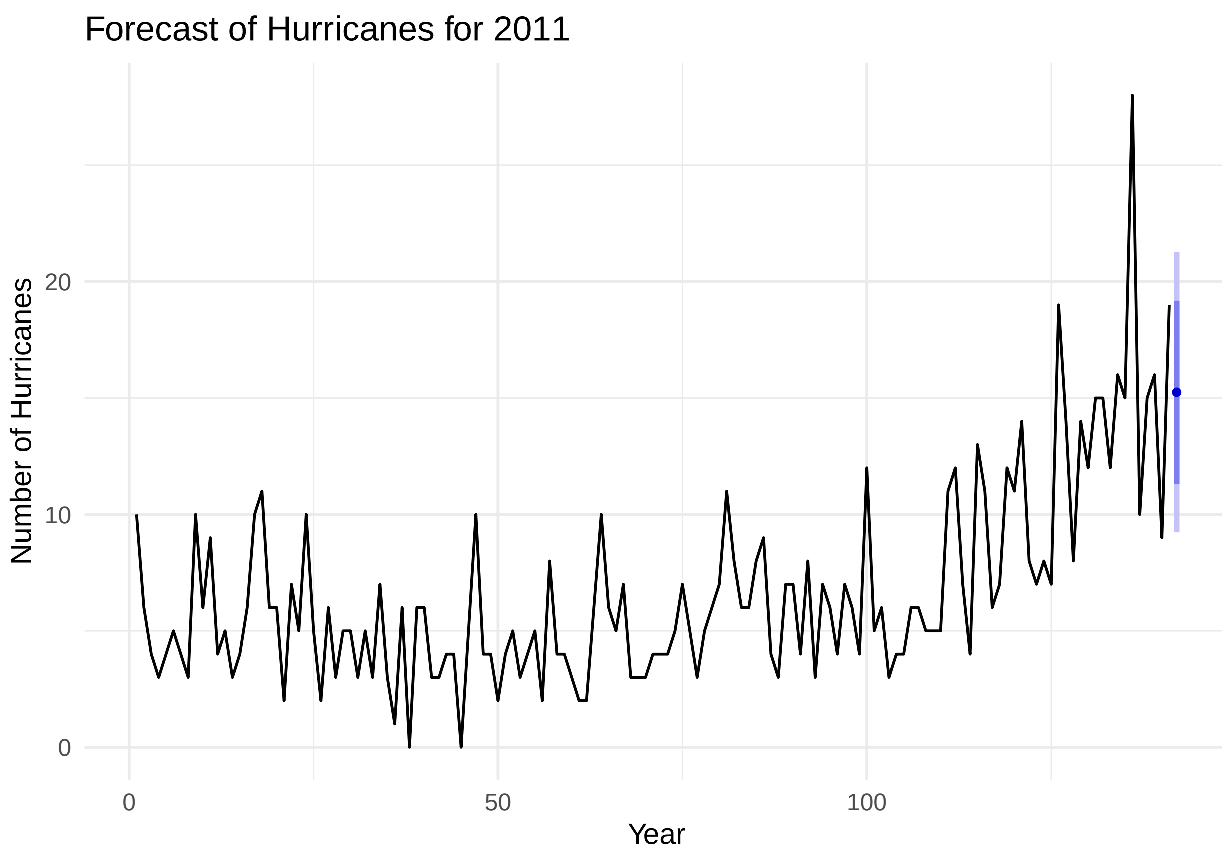
Parameter	Estimate	Std. Error
θ_1 (MA1)	**−0.8341**	0.0395
c (Drift)	**0.0694**	0.0443

4.6. Provide a forecast plot for the model using R, use the function forecast(auto.arima()) to forecast the hurricanes. Do you believe that there will be more hurricanes in 2011 year. Approximately how many more/less hurricanes do you expect as a percentage for 2011in comparison to 2010?

```
# Fit the model
model_auto <- auto.arima(timedata$Number_of_Hur)

# Forecast next 1 year (e.g., 12 months if monthly data or 1 period if yearly data)
fc <- forecast(model_auto, h = 1)

# Plot the forecast
autoplot(fc) +
  ggtitle("Forecast of Hurricanes for 2011") +
  xlab("Year") + ylab("Number of Hurricanes")+
  theme_minimal()
```




```
predicted_2011 <- 15.24614
actual_2010 <- 19
percent_change <- 100 * (predicted_2011 - actual_2010) / actual_2010

paste("The number of hurricanes expected in 2011 is", round(percent_change,2),"%")
```

```
## [1] "The number of hurricanes expected in 2011 is -19.76 %"
```